

SchoolLink

Discovery & Requirements Document

Sri Lanka School Communication Platform

1. Current Problem and Analysis

1.1 Overview of the Communication Challenge

Sri Lankan schools operate in a complex communication environment involving thousands of parents, hundreds of teachers, administrative staff, and students spread across classes, grades, and sometimes multiple schools. Today, the dominant tools used for school communication include WhatsApp groups, printed circulars, verbal relay through students, phone calls, and occasionally email. While these tools are familiar and low-cost to start, they are fundamentally ill-suited to scale and to the structured nature of school administration.

The core issue is fragmentation. There is no single authoritative source of truth for school announcements, homework deadlines, event schedules, consent requirements, or student health records. Information travels through informal, unmonitored, and often unreliable channels, leading to a cascade of downstream problems for every stakeholder group.

1.2 Key Problem Areas

1.2.1 Parents

Parents face the most visible and emotionally significant communication failures. In a typical WhatsApp group of 50 to 80 parents, important messages from teachers are buried under unrelated conversations, acknowledgements, and emoji reactions. A notice about a school medical examination, a uniform change, or a deadline for submitting a travel consent form can scroll out of view within hours.

Parents with multiple children in the same or different schools must simultaneously monitor multiple WhatsApp groups, printed diaries, and their own memory. A parent who works long hours or has low digital literacy is significantly disadvantaged. In practice, many parents only discover missed deadlines when their child is affected. A late consent form means the student cannot attend an excursion, or a missed fee reminder leads to unnecessary administrative tension.

Parent Pain Point	Impact
Missed announcements in WhatsApp groups	Children miss events, deadlines, and important notices
No visibility of homework or assignments	Parents cannot support their child's academic routine

Paper consent forms lost in transit	Children excluded from activities; admin re-sends forms
Multiple channels to monitor	Cognitive overload, especially for parents with 2+ children
No confirmation of delivery	Parents unsure if the school received a response or form
Language barriers in digital tools	Non-English or non-Sinhala parents miss critical updates

1.2.2 Teachers

Teachers carry a disproportionate share of the communication burden. A class teacher managing 35 to 45 students is simultaneously responsible for posting homework to a WhatsApp group, following up on missing consent forms, reminding parents of payment deadlines, responding to individual parent messages across multiple chat threads, and maintaining their own records of who has submitted what.

This workload compounds across subjects. A student may have five or six subject teachers, each posting in different groups or communicating through different channels. There is no consolidated view for any teacher of the total communication workload placed on a student or parent within a single school day.

Critically, teachers receive no confirmation receipts. A homework posted in a group chat may be seen by 40 parents or 10. There is no way to know. This forces teachers to repeat reminders, which further clogs communication channels and reduces the signal-to-noise ratio.

1.2.3 School Administrators

School administrators face structural inefficiency. Official notices fee payment reminders, health check schedules, sports day confirmations, term-end dates and they are drafted by the administration and then distributed through teachers to parents, typically via WhatsApp or printed circulars. This chain introduces delays, inconsistencies, and no mechanism for tracking acknowledgement or response rates.

Consent and permission form management is a particular pain point. Paper forms travel from the school to students to parents and back, with inevitable loss, delay, and incomplete returns. Chasing up missing forms is time-consuming and administratively costly. There is also no searchable archive of past notices, decisions, or responses and institutional memory exists only in individual teachers' notebooks or personal WhatsApp histories.

1.2.4 Students

Students, particularly at secondary level, are placed in an ambiguous role. They are the physical carriers of information between home and school and receiving printed notices, relaying verbal messages, and passing consent forms back and forth. This creates an obvious dependency, if the student loses a form, forgets a message, or is absent on the day a notice was distributed, the communication chain breaks entirely.

Students also lack structured visibility into their own academic obligations. In the absence of a digital assignment tracker, a student must rely on their own notes, their peers, or memory to track deadlines across subjects. Students who are disorganised or under-supported at home are the most likely to fall through these gaps.

1.3 Root Cause Analysis

The problems described above share a common root, there is no structured, role-aware, auditable communication platform designed for the Sri Lankan school context. The tools currently in use — primarily WhatsApp, physical diaries, and printed notices. They lack:

- Role-based access and visibility (a parent should only see their own child's information)
- Delivery confirmation and read receipts for official communications
- Structured data formats for assignments, consent forms, and events
- A searchable record of past communications and decisions
- Notification management that distinguishes urgent from routine messages
- Multi-language support appropriate to Sri Lanka's linguistic diversity
- An interface accessible on low-cost Android devices with variable connectivity

1.4 Contextual Factors Unique to Sri Lanka

Any solution must account for the specific realities of Sri Lankan schools, which differ materially from the other global contexts.

Factor	Implication for Design
Language diversity	Platform must support Sinhala, Tamil, and English. Language preference per user, not per school.
Device constraints	Must perform well on entry-level Android devices (2GB RAM, Android 8+). No heavy frameworks.

Connectivity variation	Key information must load with low data usage. Offline-first approach for core features.
Digital literacy gap	UI must be intuitive for parents with limited smartphone experience. Icon-driven, minimal text UI.
Multi-child families	A parent may have children in multiple classes or schools. Unified parent view is essential.
Privacy sensitivity	Student health, family contacts, and academic records are sensitive. Data must be protected.
Multiple guardians	A student may have two or more guardians with different communication needs and access levels.
Slow or intermittent connectivity	Many parents and school staff experience unstable or slow mobile data connections. The app must handle dropped connections gracefully, queue actions locally when offline, and sync automatically when the connection is restored — without losing data or requiring the user to repeat actions.

1.5 Why a Dedicated Platform is Needed

Problem Statement

Sri Lankan schools need a structured, role-aware, mobile-first communication platform that replaces fragmented channels with a single trusted source of information for schools, teachers, parents, and students. Designed for local language, device, and connectivity realities.

2. Product Name Options and Recommendation

The current working name is SchoolLink. This section proposes three alternative names, evaluates each across five criteria relevant to the Sri Lankan school context, and provides a final recommendation.

2.1 Option 1 — Vidya Connect

Name	Vidya Connect
Script forms	Sinhala: විද්‍යා කනෙක්ට් Tamil: வித்யா கனெக்ட்

Meaning	"Vidya" (Sanskrit origin, used in both Sinhala and Tamil) means knowledge or education. "Connect" communicates platform purpose directly.
Audience fit	Excellent. Both words are instantly understood by teachers, parents, and educated students. No translation needed.
Cultural alignment	Strong. Vidya is used in school names, educational institutions, and common conversation across both Sinhala and Tamil communities.
School suitability	Very good. Sounds academic and purposeful without being bureaucratic.
Brand potential	Good. Clean, two-word name. Easy to say, easy to write, easy to remember.
Weakness	"Connect" is a common suffix in app names globally (ServiceConnect, ParentConnect, etc.), which may reduce distinctiveness internationally. Within Sri Lanka this is not a significant issue.

2.2 Option 2 — Vidula Connect

Name	Vidula Connect
Script forms	Sinhala: විදුල කනෙක්ට් Tamil: விதுல கனெக்ட்
Meaning	"Vidula" is a creative variation inspired by "Vidya" — it evokes the same educational connotation while offering a more unique and elegant sound profile.
Audience fit	Moderate. Younger teachers and parents may appreciate the modern sound, but older users or those less familiar with the origin may need explanation.
Cultural alignment	Moderate. The connection to Vidya is clear to those who know it, but is not self-evident from the word alone.
School suitability	Good. Sounds contemporary and clean, suitable for a school-facing product.
Brand potential	High. The uniqueness of the name makes it more brandable and reduces the risk of confusion with other products.
Weakness	The meaning is not immediately obvious to all users. Requires a brief explanation during onboarding or marketing. May be less effective for low-digital-literacy users.

2.3 Option 3 — Widupiyasa (විදුපියස)

Name	Widupiyasa (විදුපියස)
Script forms	Sinhala: විදුපියස Tamil transliteration: விதுப்பியஸ
Meaning	"Widupiyasa" means "a place of knowledge" or "an educational environment where learning takes place" — a direct representation of a school as a knowledge space.
Audience fit	Excellent for Sinhala-speaking communities. The word carries depth and institutional gravitas that resonates with teachers and school administrators.
Cultural alignment	Very strong. This is a purely Sinhala name rooted in local educational culture. It signals that the product was designed for Sri Lanka.
School suitability	Excellent. Formal, respectful, and clearly educational in tone.
Brand potential	Moderate. The name is distinctive and memorable, but may be harder to render in English marketing materials or app stores. Tamil users may find the name unfamiliar.
Weakness	The Tamil transliteration is awkward. The name may feel exclusionary to Tamil-medium schools. Longer than ideal for app icons, login screens, and UI labels.

3. Initial MVP — In-Scope Features

The following modules are proposed for inclusion in the first version (MVP) of the platform. These were selected based on their direct impact on the core communication problems identified.

3.1 Communication Hub

The Communication Hub is the central feature of the MVP. It provides a structured, role-aware replacement for WhatsApp groups and printed circulars.

- School-wide announcements posted by administrators and visible to all parents and teachers and students
- Class-level messages sent by teachers to the parents and students in their specific class community.
- Controlled one-to-one messaging between teachers and individual parents- **Message Queue with Priority**(for parents)
- Delivery confirmation, the system records when a parent has read a message
- Message archive, all communications are stored and searchable by date and sender

3.2 Assignment and Task Tracking

A structured homework and assignment visibility layer that replaces verbal relay and informal chat posts.

- Teachers post assignments with title, description, subject, deadline, and class
- Parents see only their own child's assignment feed
- Students also can see their own pending and completed tasks
- Submission status tracking: Submitted / Late / Missing
- Automated reminders sent to parents 24 hours before a deadline
- Future integration hook for LMS platforms (manual entry for MVP)

3.3 Smart Notification System

A configurable, intelligent notification layer that eliminates the spam problem of current WhatsApp-based communication.

- Weekly digest: a single summary of upcoming assignments and events sent every Sunday evening
- Risk alerts: targeted notification when a specific student has overdue assignments
- Event reminders sent 3 days and 1 day before scheduled events
- Notification log, parents can review missed notifications at any time
- Payment confirmation notifications

3.4 Digital Consent and Admin Forms

Schools regularly need parents to read, agree to, and return forms for field trips, medical permissions, annual data updates, photography consent, sports day participation, and more. Currently this process relies on paper forms carried by students, which leads to lost forms, late returns, and significant admin follow-up work.

This module replaces the entire paper form process with a digital workflow inside the app.

How it works:

A school administrator creates a form inside the platform by selecting the fields needed such as Yes/No agreement, a text response, a date, or a file upload. The form is then sent directly to the relevant parents (by class, grade, or school-wide) with a defined deadline. Parents receive a notification, open the form in the app, fill it in, and submit it with a digital acknowledgement or finger signature.

What the admin sees:

The admin dashboard shows a live response summary at all times, how many parents have submitted, how many are still pending, and how many are overdue. The system automatically sends reminder notifications to parents who have not responded, three days and one day before the deadline. Once the deadline passes, all responses are stored in an archive that can be exported as an Excel file for school records.

Ex:

- Field trip and excursion consent
- Medical treatment permission
- Annual family contact and address confirmation
- Sports day and inter-school event participation
- Photography and media consent
- School fee acknowledgement
- Travel declarations

3.5 Unified School Calendar and Event management

A structured, role-aware school event calendar that replaces scattered date announcements.

- School administrators post events (term dates, sports days, exams, meetings) to the calendar

- Teachers can add class-specific events (tests, project deadlines, events)
- Parents see a unified calendar combining school events and their child's class events
- Export to Google Calendar and Apple Calendar for personal device integration
- Upcoming events are highlighted on the parent dashboard home screen

3.6 Student Health Module

A secure, role-restricted health record module that stores essential student health information. This information is critical in emergency situations and reduces the administrative burden of collecting health data each academic year.

- Medical details: blood type, known allergies, chronic conditions, emergency medication
- Vaccination record: vaccines administered with dates and provider
- Annual checkup record: height, weight, BMI, vision and dental checkup results
- Health tips library: curated, age-appropriate health tips linked from trusted sources (Ministry of Health, WHO)
- Emergency access: in emergency, school nurse or designated admin can access core health data for a student
- Privacy controls: health data visible only to designated admin roles and the parent not to teachers by default

3.7 Insights Dashboard

A summary dashboard for each user role, showing the most important information at a glance.

- Parent dashboard: upcoming assignments, unread messages, pending consent forms, next events
- Teacher dashboard: class assignments submitted vs outstanding, unread parent messages, upcoming class events
- Admin dashboard: school-wide announcement reach, outstanding consent forms, health record completeness

3.8 Discipline Module

- Record positive behaviour (merits, commendations) and negative behaviour (demerits, warnings, suspensions)
- Attach notes and supporting details to each incident
- Notify parents instantly via push notification or SMS when an incident is recorded
- Parents can view full discipline history but cannot edit or dispute records within the app
- Teachers can notify admin of conflicts or escalate incidents with one tap

- Discipline history feeds directly into character certificate generation, removing the need for manual drafting

3.8 Bulk Data Management

A data management capability to reduce the manual effort of setting up and maintaining the platform, particularly for school administrators with limited time and technical resources.

- Bulk upload of student information, parent details, and class assignments via Excel templates
- Bulk creation of parent logins, student logins, and staff logins from uploaded data
- Bulk import of exam marks and academic records using standardised Excel sheet formats
- Downloadable template files for each upload type to ensure data consistency
- Validation and error reporting during upload: rows with errors are flagged with clear messages
- Bulk export of any module's data (students, assignments, consent responses) for school records

4. Out-of-Scope Features (Post-MVP)

They are valuable and worth building in future versions, but they introduce complexity that would delay the first release and distract from the core communication and health problems the MVP is designed to solve.

4.1 AI Assistant

A conversational AI assistant integrated into the parent and teacher interfaces.

- Summarise a student's weekly activity in plain language on request
- Answer parent questions such as "Are there any assignments due this week?" or "When is the next parent-teacher meeting?"
- Proactively highlight academic risk signals to parents before they escalate
- Automate routine teacher communications such as weekly progress summaries

This feature is powerful and differentiating, but requires API integration, prompt engineering, and careful cost management. It is best added in Version 2 once the platform has stable data to work from.

4.2 Library Management Module

A complete school library management system integrated into the platform.

- Track and display physical book location using barcode or QR code scanning
- Student and management book reservation system
- Book search by category, ISBN, serial number, author, title, and publisher
- Bulk upload of library catalogue from Excel sheet
- Customised report generation: library inventory, book summaries, due date reports, unavailable books, checkout details
- Automatic fine calculation for late returns based on configurable rate and grace period
- Automated book due alerts sent to students and parents before and on the return date

Library management is a well-defined and valuable module, but it serves a distinct use case from school communication. It is best built as a standalone sub-module in a later version after core communication is stable.

4.3 Alumni Module

A dedicated alumni management and engagement module for past students.

- Create and manage alumni profiles from past student records already in the system
- Post and manage alumni news, achievements, and events in the school communities
- Organize and promote fundraising campaigns for the school

- Issue alumni login credentials from existing student data with minimal admin effort
- Alumni networking features: class reunion coordination, achievement spotlights

The alumni module is a long-term community-building feature. It requires a matured user base and active school engagement. It is more relevant after the platform has been in use for several years and the first cohort of student users have graduated.

4.4 LMS Integration

- Integration with existing Learning Management Systems (e.g. Google Classroom, Moodle) to automatically pull assignment and homework data into the platform rather than requiring manual teacher entry.

5. Initial User Role List with Functional and Non-Functional Requirements

5.1 Role Overview

Role	Access Scope	Primary Goal
Super Admin	Full system	Manage the platform, configure schools, manage all data
School Admin	School-wide	Manage school data, post notices, run consent forms, oversee all classes
Teacher	Assigned class(es)	Post assignments, communicate with class parents, view class-specific reports
Parent / Guardian	Own child(ren) only	Receive updates, respond to forms, track child's academic progress
Student	Own record only (read)	View assignments, deadlines, and optionally health summary

5.2 School Administrator — Requirements

Functional Requirements

- Create and manage school profile, academic year, and term dates
- Create, edit, and deactivate teacher, parent, and student accounts
- Post school-wide announcements with priority levels (normal, urgent)
- Create and distribute digital consent and permission forms to selected parents
- Monitor form response status and send automated reminders to non-respondents

- Bulk import student, parent, and staff data via Excel templates
- Export consent form responses, student lists, and health summaries
- View school-wide dashboard: announcement reach, outstanding forms, assignment completion rates
- Configure school calendar and add school events visible to all users
- Manage health module visibility and emergency health access permissions

5.3 Teacher — Requirements

Functional Requirements

- View assigned class roster and parent contact information
- Post assignments to assigned class with title, subject, description, deadline, and attachments
- Edit or cancel posted assignments with notification to affected parents
- Send class-level messages to all parents of an assigned class
- Send individual messages to a specific parent with message threading
- View delivery status of messages (sent, delivered, read)
- Mark assignments as graded or add feedback notes (optional in MVP)
- Add class events to the school calendar
- View class-level assignment completion rate dashboard

5.4 Parent / Guardian — Requirements

Functional Requirements

- Receive and read school-wide and class-level announcements
- View the complete assignment and homework feed for each of their children
- Receive and configure smart notifications: weekly digest, urgent alerts, event reminders
- Complete and submit digital consent and permission forms
- View and update their child's health record (medical details, vaccinations, checkups)
- View their child's health growth chart (height, weight, BMI over time)
- View the unified school and class calendar
- Send messages to their child's teacher (controlled, not group chat)
- Manage notification preferences per child
- Switch between multiple children if they have more than one enrolled

Non-Functional Requirements

- The platform must be available in Sinhala, Tamil, and English with per-user language preference

5.5 Student — Requirements (Limited Access)

Functional Requirements

- View own pending and submitted assignments across all subjects
- View upcoming deadlines on a personal calendar view
- View own health summary (height, weight, BMI chart — read-only)
- View school and class announcements relevant to their class

Non-Functional Requirements

- Student access must be limited to read-only. no ability to post, message, or modify any data
- Student login credentials must be separate from parent credentials
- Student access can be disabled school-wide at the administrator's discretion

6. Proposed Technology Stack

The following technology choices are proposed based on the project constraints: mobile-first delivery, Sri Lankan device and connectivity realities, a student team, and a need for rapid MVP development.

6.1 Mobile Application

Layer	Technology	Rationale
Mobile Framework	React Native	Single codebase for Android and iOS. Expo simplifies build and deployment. Strong community support. Familiar for web developers.
UI Component Library	React Native Paper	Material Design 3 components. Accessible, well-tested, and visually consistent across Android versions.
Offline Support	WatermelonDB	Efficient offline-first database for React Native. Allows key features to work without internet, then sync when connected.
Push Notifications	Firebase Cloud Messaging	Free tier sufficient for MVP. Supports Android and iOS. Integrates well with React Native Expo.

6.2 Web Application(Only Admin and teachers)

Layer	Technology	Rationale
Web Framework	React.js (Next.js)	Fast, SEO-friendly web framework. Same React knowledge as the mobile app reduces learning curve for the student team.
UI Component Library	Tailwind CSS	Utility-first CSS framework for rapid UI development. Clean and responsive admin interface without writing custom CSS.
Charts and Analytics	Recharts	Simple responsive charts for attendance trends, assignment completion rates, and discipline dashboards.

Form Handling	React Hook Form	Lightweight form management with built-in validation. Used for consent form builder and admin configuration screens.
Excel Export	SheetJS	Generate and download Excel reports directly from the browser.

6.3 Backend and Database

Layer	Technology	Rationale
API Framework	Node.js + Express.js	Widely taught, large ecosystem, fast to prototype. REST API suits mobile client needs well.
Primary Database	PostgreSQL	Relational model suits structured school data (students, classes, assignments). Strong JSON support for flexible fields.
Authentication	JWT auth with role support	Stateless auth suitable for mobile. Role claims in token enable role-based access control without extra DB calls.
File Storage	Supabase Storage Or AWS S3(need mentor feedback)	Natively integrated with the Supabase backend, eliminating the need for a separate storage service. Supports role-based access control, ensuring sensitive files such as health records are only accessible to authorised users. Free tier is sufficient for pilot school deployment and scales to production without infrastructure changes.
Excel Processing	ExcelJS (Node)	Read and write Excel files server-side. Supports bulk data import validation and export with formatting.

6.4 Infrastructure and DevOps

Layer	Technology	Rationale
Hosting	Railway or Render (cloud PaaS)	Zero-config deployment for Node + Postgres. Free tier suitable for MVP. No DevOps overhead for student team.
CI/CD	GitHub Actions	Automated test and deploy on push. Free for public and student repositories.
Version Control	Git + GitHub	Industry standard. Enables code review, branching strategy, and team collaboration.

7. Assumptions Requiring Mentor Feedback

7.1 Technical and Design Questions

- We have included WatermelonDB for offline support. If connectivity at pilot schools is reliable, offline sync can be simplified to reduce development time.
- We have assumed parents or health center staff enter and own their child's health data, with school admin having read access for emergencies.
- We have assumed best-practice data privacy aligned with GDPR principles, as Sri Lanka does not yet have a fully enforced personal data protection law. This affects data storage, consent flows, and deletion rights.

7.2 Process and Delivery Questions

- Q1— What is the way the system can earn revenue?

Suggestions

- Schools pay a monthly or annual fee based on their student count. A small school under 300 students pays a low flat rate. A large school above 800 students pays a higher tier.